Spectral analysis and block convolution

Practice questions · Day 11 · Tue 3 Nov · pack D6

5. Practice

From your course files

T5 Q5 (tutorial). Let $x_a(t)$ be an analogue signal with bandwidth $B = 3$ kHz. We wish to use a $N = 2^m$-point DFT to compute the spectrum of the signal with a resolution less than or equal to 50 Hz. Determine
(a) the minimum sampling rate,
(b) the minimum number of required samples
(c) the minimum length of the analogue signal record.

T5 Q4 (tutorial). Consider a causal LTI system with transfer function

$$H(z) = \frac{1}{1 - 0.5z^{-1}}.$$

The output signal of the system, $y(n)$, is known for $0 \leq n \leq 63$. Assuming that $H(z)$ is also known, develop a method to recover the input signal $x(n)$ for $0 \leq n \leq 63$ using the 64-point DFT and IDFT.

T5 Q6 (tutorial).
(a) Determine the discrete-time Fourier transform $X(\omega)$ of the signal $x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$.
(b) Compute the 6-point DFT $V(k)$ of the signal $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.
(c) What is the relationship between $X(\omega)$ and $V(k)$? Explain.

New questions

N1. [3 marks] A signal is sampled at $F_s = 1$ kHz and analysed with a $N = 200$ point DFT (no zero-padding).

a. What is the frequency resolution and the record length? [1]
b. A real tone at 120 Hz is present. In which bins does it appear? [1]
c. What analogue frequency does bin $k = 37$ represent? What about $k = 150$? [1]

N2. [4 marks] $L = 100$ samples of a signal are taken at $F_s = 1$ kHz and zero-padded to $N = 512$.

a. Find the bin spacing and the frequency resolution. [2]
b. The signal contains tones at 200 Hz and 205 Hz. Can this DFT separate them? Would zero-padding to 4096 help? [1]
c. What is the minimum number of samples needed to separate them, and the matching power-of-2 DFT size without padding? [1]

N3. [5 marks] An FIR filter $h(n) = \{\underset{\uparrow}{1}, -1\}$ is applied to $x(n) = \{\underset{\uparrow}{1}, 0, 2, 1, 3, 1\}$ using the overlap-add method with blocks of $L = 3$ input samples.

a. What DFT length N is needed? Write out the padded $h(n)$. [1]
b. Find the output of each block. [2]
c. Combine them to get $y(n)$ and check one value directly. [2]

N4. [8 marks] The filter $h(n) = \{\underset{\uparrow}{2}, -1, 1\}$ is applied to $x(n) = \{\underset{\uparrow}{1}, 3, 2, 1, 0, 2, 4, 1\}$ using the overlap-save method with $N = 5$ point DFTs.

a. How many new input samples does each block carry? [1]
b. Write out every input block needed to produce the complete output. How many blocks are there? [2]
c. Compute the 5-point circular convolution of h with the first two blocks. Say which outputs you discard and why. [3]
d. Give the complete output $y(n)$. How many blocks would overlap-add have needed with the same N? [2]

N5. [5 marks] A signal sampled at $F_s = 8$ kHz contains a strong tone at 1010 Hz (amplitude 1) and a weak tone at 1050 Hz (amplitude 0.01, i.e. -40 dB). You are told that the worst-case leakage of a tone into a bin d or more bins away is:

d (bins)	2	3	4	5	6
Rectangular	-18 dB	-21 dB	-23 dB	-25 dB	-26 dB
Hanning	-31 dB	-41 dB	-48 dB	-54 dB	-58 dB

a. Which window must be used, and why? [2]
b. Find the minimum power-of-2 record length L (with $N = L$) so that the weak tone is at least 5 bins from the strong one. Give T_0. [2]
c. What would you see with a rectangular window and the same L? [1]

N6. [9 marks] An analogue signal band-limited to 5 kHz is sampled by an ADC fixed at $F_s = 12$ kHz. It contains two equal-amplitude tones at 1500 Hz and 1530 Hz.

a. Is F_s adequate? [1]

b. Using a rectangular window and $N = L = 2^m$, find the minimum N that resolves the tones, the record length and the actual resolution. [2]
c. For that N, find the bin index of each tone. Comment on leakage. [2]
d. A colleague records only 128 samples and zero-pads them to 2048. Find the bin spacing and the resolution. Will the tones appear as two peaks? [2]
e. The colleague then insists on a Hanning window. Using the rule of thumb that Hanning needs about twice the record length, find the new minimum power-of-2 L and T_0. [2]

6. Quick recall check

1. $F_s = 2$ kHz, $N = 256$, no zero-padding. What is the resolution, and what is T_0?
2. Does zero-padding from 256 to 1024 samples let you resolve tones that were unresolved before? What does it do?
3. Write the 5-point Hanning window from the slide 32 formula.
4. Overlap-save, $M = 4$, $N = 16$: how many new samples per block, and which outputs do you discard?
5. Give the main-lobe and side-lobe trade-off of the Hanning window compared with the rectangular one, in one sentence.